

AWS Certified AI Practitioner (AIF-C01)

Domain 1 Study Notes — Fundamentals of AI/ML, with Amazon SageMaker Deep Dive

Part 1 — What is Amazon SageMaker

Amazon SageMaker is AWS's fully managed machine learning platform. It provides the tools to prepare data, build, train, tune, deploy, and monitor ML models — all in one integrated service — so teams don't have to stitch together their own infrastructure for each stage of the ML lifecycle.

Exam framing: For AIF-C01 you are NOT tested on how to use SageMaker (no coding, no config). You ARE tested on recognizing SageMaker feature names and matching them to the ML lifecycle stage or business problem they solve. Focus on the "what does this feature do" level, not implementation.

1.1 The Three Core Stages SageMaker Covers

Stage	What Happens	Key SageMaker Feature
Build	Prepare data, explore it, engineer features, choose an algorithm	SageMaker Studio, Data Wrangler, Feature Store, Canvas
Train	Feed data to an algorithm so the model learns patterns; tune hyperparameters	Training jobs, Automatic Model Tuning, Experiments, Debugger
Deploy	Host the trained model so it can generate predictions (inference)	Real-time endpoints, Batch Transform, Serverless Inference, Model Monitor

1.2 SageMaker Features Most Likely to Appear on the Exam

SageMaker Studio

A web-based IDE that brings together all ML development tools — notebooks, data prep, training, deployment, monitoring — in a single visual interface.

SageMaker Data Wrangler

Reduces the time to aggregate and prepare data for ML from weeks to minutes. Provides a visual interface for data import, cleaning, and feature engineering with built-in transforms — no need to write custom preprocessing code.

SageMaker Feature Store

A centralized repository to store, share, and reuse engineered features across teams and models. Solves the problem of duplicated feature engineering work and inconsistent features between training and inference (training/serving skew).

SageMaker Ground Truth

A managed data labeling service. Uses human labelers (your own workforce, Mechanical Turk, or vendor workforces) plus active learning to reduce labeling cost and time — directly relevant to "labeled vs unlabeled data" concepts in Domain 1.

SageMaker Autopilot

AutoML capability — automatically explores different algorithms and hyperparameters and builds a model for you, given tabular data and a target column. Useful when the exam asks about "automating model selection."

SageMaker Canvas

A no-code visual interface aimed at business analysts. Lets non-technical users build ML models and generate predictions through point-and-click — no notebooks or code required.

Automatic Model Tuning (Hyperparameter Tuning)

Automatically searches across hyperparameter combinations (e.g., learning rate, batch size) to find the set that produces the best-performing model, using strategies like Bayesian optimization.

SageMaker Clarify

Detects potential bias in datasets and trained models (e.g., a model predicting negative outcomes more often for one demographic group) and helps explain model predictions (feature attribution / explainability). This is a Domain 4 (Responsible AI) crossover — expect it there too.

SageMaker Model Monitor

Continuously monitors deployed models in production for data drift and model quality degradation, so you know when a model needs retraining. Key term to associate: "drift detection."

SageMaker Model Cards

A single place to document model details — intended use, training data, evaluation results, risk considerations — for governance and transparency. Directly tested in Domain 4/5 under "transparent and explainable models."

SageMaker JumpStart

A hub of pre-trained, open-source foundation models and solution templates that can be deployed with a few clicks or fine-tuned on your own data. This is the bridge between Domain 1 (traditional ML) and Domain 2 (generative AI) — know that JumpStart hosts foundation models, while Bedrock provides managed API access to foundation models.

SageMaker Debugger

Provides real-time visibility into training jobs (e.g., detecting vanishing gradients, overfitting) so training issues are caught early.

SageMaker Experiments

Tracks, organizes, and compares different training runs (experiments) so you can reproduce and compare results across model iterations.

SageMaker Neo

Compiles and optimizes trained models to run efficiently on specific hardware targets (cloud instances or edge devices), improving inference performance without loss of accuracy.

SageMaker Inference Options

Inference Type	Use Case
Real-time inference	Low-latency, persistent endpoint for live predictions (e.g., a web app calling a model per request)
Batch Transform	Run predictions on a large batch of data at once, no persistent endpoint needed
Serverless Inference	Auto-scales to zero when idle — good for intermittent, unpredictable traffic

Inference Type	Use Case
Asynchronous Inference	Queues large payloads/long-running inference requests and processes them in the background

1.3 SageMaker's Relationship to Other Exam Services

Service	How it Differs from SageMaker
Amazon Bedrock	Managed access to third-party and Amazon foundation models via API — for generative AI apps. SageMaker JumpStart also hosts FMs, but Bedrock is the primary "serverless GenAI" service.
Amazon Comprehend	Purpose-built, pre-trained NLP service (sentiment, entities) — no training required, unlike SageMaker where you build/train your own model.
Amazon Rekognition	Purpose-built pre-trained computer vision service (image/video analysis) — same idea, no custom training needed.
Amazon Personalize / Forecast / Fraud Detector	Purpose-built ML services for specific business problems (recommendations, forecasting, fraud) — abstract away the ML pipeline entirely.

Exam pattern: if a question describes a generic/custom ML problem needing full control over data, algorithm, and training → answer is SageMaker. If it describes a narrow, well-known task (translate text, detect faces, transcribe audio) → answer is the purpose-built AI service (Comprehend, Rekognition, Transcribe, etc.), because it requires "the least operational overhead."

Part 2 — Domain 1: Fundamentals of AI and ML (20% of exam)

Domain 1 has three task statements. Below is what to study for each, focused on the AWS-specific angle since the general ML theory overlaps with your existing background.

Task Statement 1.1 — Explain basic AI concepts and terminologies

Core definitions to have crisp one-line answers for

Term	One-line definition
AI	Broad field: machines performing tasks that normally require human intelligence
ML	Subset of AI: systems that learn patterns from data rather than explicit rules
Deep Learning	Subset of ML using multi-layer neural networks
Neural Network	Layers of interconnected nodes (neurons) that learn weighted representations of data
Computer Vision	AI applied to interpreting images/video
NLP	AI applied to understanding/generating human language
Model	The artifact produced by training — encodes learned patterns
Algorithm	The method/procedure used to learn from data (e.g., linear regression, XGBoost)
Training	Process of fitting a model's parameters to data
Inference	Using a trained model to make predictions on new data
Bias (fairness sense)	Systematic unfairness in model outputs toward a group
Bias (statistics sense)	Error from overly simplistic assumptions (underfitting)
Fit	How well a model captures patterns — underfit, good fit, or overfit
LLM	Large Language Model — a foundation model trained on massive text corpora

Inferencing types

- Real-time inference: low-latency, on-demand predictions (e.g., fraud check during checkout)
- Batch inference: predictions run on large datasets on a schedule, not latency-sensitive (e.g., nightly churn scoring)

Data types to recognize

- Labeled vs unlabeled data — labeled has known target outputs (supervised learning); unlabeled does not (unsupervised learning)
- Tabular — rows/columns, structured (e.g., spreadsheets, database tables)
- Time-series — sequential data indexed by time (e.g., stock prices, sensor readings)
- Image — pixel data, used in computer vision
- Text — unstructured natural language
- Structured vs unstructured — structured fits neatly into rows/columns; unstructured does not (text, images, audio)

Learning paradigms

- Supervised learning — learns from labeled data; used for classification and regression

- Unsupervised learning — finds patterns in unlabeled data; used for clustering and anomaly detection
- Reinforcement learning — learns via trial and error using rewards and penalties from an environment

Task Statement 1.2 — Identify practical use cases for AI

When AI/ML adds value

- Assisting human decision-making (e.g., recommendation engines)
- Scaling solutions beyond what manual rules can handle
- Automating repetitive pattern-recognition tasks

When AI/ML is NOT appropriate

- When a deterministic, guaranteed outcome is required instead of a probabilistic prediction (e.g., calculating tax owed — use a formula, not an ML model)
- When the cost of building/maintaining an ML solution outweighs its benefit (cost-benefit analysis)
- When there isn't enough quality data to train a reliable model

Matching ML technique to use case

Technique	Use When
Regression	Predicting a continuous numeric value (e.g., house price)
Classification	Predicting a category/label (e.g., spam vs not spam)
Clustering	Grouping similar unlabeled data points (e.g., customer segmentation)

Real-world AI application categories to recognize

- Computer vision — Amazon Rekognition
- NLP — Amazon Comprehend
- Speech recognition — Amazon Transcribe
- Speech synthesis — Amazon Polly
- Translation — Amazon Translate
- Chatbots/conversational AI — Amazon Lex
- Recommendation systems — Amazon Personalize
- Fraud detection — Amazon Fraud Detector
- Forecasting — Amazon Forecast

Know at a glance: purpose-built AWS AI services (no training required)

Service	What it Does
Amazon Transcribe	Speech-to-text
Amazon Translate	Language translation
Amazon Comprehend	NLP — sentiment, entity/keyphrase extraction, PII detection
Amazon Lex	Build conversational chatbots/voice bots (same tech as Alexa)
Amazon Polly	Text-to-speech
Amazon Rekognition	Image and video analysis (object/face detection, moderation)
Amazon Textract	Extract text and data from scanned documents/forms

Service	What it Does
Amazon Kendra	Intelligent enterprise search
Amazon Personalize	Real-time personalized recommendations
Amazon Fraud Detector	Detect potentially fraudulent online activity

Task Statement 1.3 — Describe the ML development lifecycle

Pipeline stages (know this order)

- 1. Data collection
- 2. Exploratory Data Analysis (EDA)
- 3. Data pre-processing (cleaning, transforming)
- 4. Feature engineering
- 5. Model training
- 6. Hyperparameter tuning
- 7. Evaluation
- 8. Deployment
- 9. Monitoring (and retraining as drift occurs — the cycle repeats)

Sources of ML models

- Open-source pre-trained models (e.g., via SageMaker JumpStart)
- Training a custom model from scratch on your own data

Methods to use a model in production

- Managed API service (e.g., calling Amazon Bedrock or a SageMaker endpoint)
- Self-hosted API (you deploy and manage the serving infrastructure yourself)

MLOps fundamentals

- Experimentation — trying different models/parameters in a tracked, reproducible way
- Repeatable processes — pipelines that can be re-run reliably
- Scalable systems — infrastructure that grows with data/traffic
- Managing technical debt — keeping pipelines maintainable over time
- Production readiness — testing, validation, and governance before go-live
- Model monitoring — tracking live performance for drift
- Model re-training — updating models as new data becomes available

Model performance metrics vs business metrics

Type	Examples	What it Measures
Model performance metrics	Accuracy, AUC (Area Under ROC Curve), F1 score	How well the model performs the ML task itself
Business metrics	Cost per user, development cost, customer feedback, ROI	Whether the model delivers real business value

Exam tip: A question may describe great model accuracy but poor ROI (or vice versa) — the "correct" answer usually weighs business value, not just the technical metric, since AWS frames this exam around practical business application.

Relevant SageMaker features per pipeline stage (cross-reference to Part 1)

Pipeline Stage	SageMaker Feature
Data collection / prep	Data Wrangler, Ground Truth (labeling)
Feature engineering	Feature Store
Training	Training jobs, Experiments, Debugger
Hyperparameter tuning	Automatic Model Tuning
Evaluation	Clarify (bias), Experiments (comparison)
Deployment	Real-time / Batch / Serverless / Async inference
Monitoring	Model Monitor
Governance/documentation	Model Cards

Sources: AWS official AIF-C01 Exam Guide (v1.4); AWS SageMaker documentation (docs.aws.amazon.com/sagemaker).